

PROJECT SUMMARY

This project is designed to teach the fundamental principles of mobile robotics and enhance mechatronic integration skills. The robot features a multi-layer hardware architecture equipped with environmental perception sensors. Low-level control is managed by an Arduino Mega 2560, while high-level processing is executed on a Raspberry Pi 3B+.

The main objective is to integrate perception, action, and decision-making processes to develop a mobile platform capable of autonomous movement.

DESIGN AND MORPHOLOGY

The robot's structure is built on a three-layer architecture:

Base Layer: Motor drivers, battery, motors, wheels, and sensor connections.

Middle Layer: Arduino Mega 2560, Raspberry Pi 3B+, TB6612FNG motor driver, IR sensors, and LIDAR module connections.

Top Layer: YLIDAR 360, DC-DC converter, and wiring layout.

This design ensures balanced weight distribution and easy maintenance access.

HARDWARE COMPONENTS

The main hardware components used in the project are:

Sensors:

- 3× Sharp GP2Y0A21YK0F IR sensors (10–80 cm range)
- YLIDAR 360° LIDAR sensor
- 298:1 micro metal DC motors with encoders

Control Units:

- Arduino Mega 2560 (low-level control)
- Raspberry Pi 3B+ (high-level processing and ROS integration)

Drivers and Power Elements:

- TB6612FNG dual DC motor driver
- 7.4V Li-ion battery
- DC-DC converter (5V USB output)

Additional Parts:

- RGB LED module, caster wheels, mounting components, and a custom 3-layer chassis.

SOFTWARE AND DECISION-MAKING ARCHITECTURE

The robot's control system consists of two main levels:

- Low Level (Arduino Mega): Motor control, encoder data reading, and sensor data processing.
- High Level (Raspberry Pi – ROS): Mapping, navigation, decision-making, and basic artificial intelligence algorithms.

This architecture allows the robot to perceive its surroundings and perform tasks autonomously.

